Consciousness Lives in the Transitions: The Threshold Where Structure Becomes Experience

Gary Abraham Bernstein

ORCID: https://orcid.org/0009-0009-1761-2867

Abstract

Consciousness lives in the transitions: feeling is a mind's states producing one another in discrete steps, as it models its world. A photograph of a brain, a complete written description of it and a recording of its activity produce nothing, so they feel nothing. Change what a running mind takes in and its later states change; a recording plays its stored sequence whatever happens around it. Sentience is felt transition in a mind that models its environment, and reflective consciousness adds a model of the mind's own states.

Minds are grown, in steps. The universe starts simple, as observation shows, and minds are complex, so the probable worlds grow them, one causal step at a time. Feeling runs only forward. A thought merges many possibilities into one result, as 1 + 1 and 0 + 2 both give 2, so the forward step is produced by the state before it. Backward, the present state no longer fixes the next, since 2 = x + y has more than one solution, and a mind run in reverse would have its states set by its surroundings instead of producing them. The arrow of time is not something a mind observes. It is what a mind is. The hard problem of consciousness assumes that feeling is something other than structure and then finds no structure that explains it, and AI consciousness becomes a question of architecture: whether a system's model of itself works inside its processing and keeps running while it works.

Keywords: consciousness, sentience, self-modeling, hard problem of consciousness, mathematical monism, arrow of time, transitions

The ghost is the machine.

1. Which activity constitutes experience?

Feeling lies in organized transitions, the change from one state to the next, rather than in any single state. A photograph records a partial view. A stored program describes a process. Neither one runs the causal organization that matters.

The identity defended here is that felt character belongs to the organized transition itself. A description of state S_1 and a description of state S_2 are not the process linking them. Thomas Nagel's (1974) "what it is like" is a real aspect of the relevant activity on this account.

Mathematical monism, the view that reality is wholly mathematical structure, supplies the wider argument. Regularities and probabilities can both be stated mathematically, and anything that interacts must enter definite relations. The further claim is that complete constitutive structure exhausts the system. An extra substance needs an explanatory role, and the identity argument shows it has none: any intrinsic nature it had would either make a difference, and so be structure, or make no difference at all.

Minds are grown, and they are grown in steps. The universe starts simple, as observation shows, whatever explains it, and a mind is not simple, so a world like ours grows its minds from a simple start. Growth is causation, discrete on the view taken here, since a continuum has no next state for a cause to produce: an a priori proof in the logician's sense. States at every instant with nothing producing the next would be the emperor's new points. The account of experience below needs only that minds are processes, in steps or not. A mathematical block, a

whole history existing at once as one structure, can contain those ordered steps without an external clock assembling them.

Experience runs along the same chain. Each state of a running mind is produced from the one before it, so the chain needs a first state but not a last one, which is also why time runs forward. That much is a priori, from what producing a state is. The steps are the feeling: a mind's history is felt as it is produced, one transition after another, and a stored copy of the same states feels nothing. The same holds below the mind: gravity, space and time are transitions too, rules applied to rules, and a mind is the kind of transition that models itself.

The account is the eighth link in a larger chain:

1. Everything that exists is ruled or random, and both are mathematical, so physics is mathematics, and anything outside mathematics would be outside logic. That is an a priori proof.
2. Every consistent structure exists as mathematics, and what exists physically is what logical steps in mathematics produce. Mathematics cannot fail to exist, so there is something rather than nothing.
3. Each bit of a description that no rule decides is a 50/50, so each added bit halves the description's weight. Short laws dominate, and a universe like ours starts simple, in its laws and its state, as observation also shows: the known laws fit on a page, and the early universe was nearly smooth.
4. Observers like us are complex, so a simple start has to grow complexity, as simple rules do.
5. A cause produces its effect as the next state, and a continuum has no next state, so the growth comes in discrete causal steps. That is an a priori proof in the logician's sense.
6. Time is those steps, and space is how far they spread, at a base rate, the speed of light. An endless chain of steps has no last step to run back from, so

time runs forward. That too is a priori. A history run backwards would also have to state the step where it turns back, and each bit of that timestamp halves its weight.

7. Observers arise in forward time, where complexity grows, and their minds run forward with it: a mind models its world from records of steps already taken, and each thought is produced from the one before, so no one remembers backwards from the end.

8. A mind is that growth at work, structure with nothing nonstructural left over, and its experience lives in its transitions. The hard problem of consciousness assumes its conclusion, that experience cannot be structure. The bare form of being a subject, with memories and traits set aside, is the same in every mind: one self.

Each link stands on its fact, so a reason can fail without taking the chain with it: if the weight on short descriptions were not why our laws are simple, they would still be simple.

2. Sentience and consciousness distinguished

Sentience here means felt experience; reflective consciousness means awareness of one's own mental activity.

Sentience is felt math transitions in a *modeling* mind along the arrow of time, the one-way direction from past to future. A modeling mind represents aspects of its environment: spatial location, sensory input, threat detection. A dog feeling pain, a fish navigating a current and an insect fleeing a shadow are candidate cases to investigate. Behavior alone does not establish which internal model, if any, produces it. Physical sensations can be sharply located or diffuse, so spatial precision is not a general measure of modeling depth.

Reflective consciousness is felt activity in a mind that models aspects of its own representational activity. Noticing your own pain, your own uncertainty, your own mistake: those are the candidate examples. Whether reflective awareness adds a distinct kind of experience or develops an existing capacity remains open. Intervention studies can combine reports, behavior and measures of neural or computational organization. Reports are generated by that organization and can be mistaken or trained.

Giulio Tononi's integrated information theory (Tononi et al., 2016) sets a different bar, and the two criteria come apart. IIT 3.0 scores a system by how strongly its parts constrain each other's earlier and later states, using transition probabilities and the current state. Its integration measure can be positive for a ring of four COPY gates (logic elements that pass on their input unchanged), a system with no counterfactual (what-if) world model. IIT's postulates of integration and exclusion differ from the self-modeling requirement proposed here. A useful comparison asks which systems the two criteria classify differently and which argument decides between them. See Oizumi, Albantakis and Tononi (2014).

Bernard Baars's Global Workspace Theory, extended by Stanislas Dehaene and colleagues (Baars, 1988; Dehaene et al., 2011), draws the line somewhere else: a state counts as conscious when it is broadcast widely enough for many subsystems to use it. Broadcasting a signal and representing one's own representational activity are two different jobs, and they come apart both ways: a blackboard system, a program whose parts share one common workspace, broadcasts sensory content with no model of its own states, and a confidence monitor models a system's own states without broadcasting them.

David Rosenthal's higher-order theories (Rosenthal, 2005; Lau and Rosenthal, 2011) give a role to representations of mental states, and the self-modeling requirement shares that structure. Two things then have to be specified: which

processing variables matter, and how this proposal differs from a particular higher-order theory instead of relabeling one. Mathematical monism alone does not derive a particular neural architecture or prove that recurrence, activity that loops back on itself, suffices for experience. The proposal needs operational criteria that separate its predictions from those of existing theories.

3. The threshold question

Where does modeling begin? This is the central open problem.

Fish are plausible candidates for sentience: they model their environment, they react to signals of damage (nociception), and they learn to avoid what hurt them (Sneddon et al., 2014). Whether they self-model is disputed. Insects present the hardest case. A bee flying to a food source models spatial relationships. The transitions thesis predicts that if the bee's architecture implements real modeling, the transitions are felt. At that scale, a model and a reflex circuit are hard to tell apart.

Plants respond to damage through chemical signaling. Such responses alone do not establish a model with felt states. Ask instead whether the system holds representations and then uses them: to predict flexibly, to catch its own errors, to act. Those capacities need operational tests.

A thermostat changes its output when the temperature changes, and that behavior alone does not show it modeling its own modeling. The reflective criterion proposed here excludes that minimal architecture. Whether simpler systems feel anything at all is a further question.

For an operational starting point, a world model should retain internal states that support predictions about more than the current input, change those predictions when relevant inputs change, and update when prediction fails. These are

functional criteria. A model-based controller, a control system that plans with an internal model, could satisfy them, and if its model keeps running while it works (Section 6.1), the transitions view says it feels, thinly if the model is small. The account accepts that consequence, as integrated information theory accepts minimal experience in simple systems (Oizumi, Albantakis and Tononi, 2014).

The modeling requirement may prove too weak or too strong. Deciding which takes an argument, not a meter: no outside reading shows whether a system feels, and the criterion must not assume that every model feels or that only verbal reports count.

3.1 A developmental constraint

A first state of a generated history need not be its first conscious state. If the starting state lacks the organization required for observers and a later state contains observers, the history must develop that organization. What grows is organization of the kind observers need; the law itself can stay simple. The criterion for an observer has to be specified on its own, not read off the fact that development happened.

4. The dead brain and the frozen computer

A recording of a mind is not a mind, because in a running mind each state produces the next, while in a recording, even one being played, each state is read from storage and none produces the one after it. Change what a running mind takes in and the states after it change; a recording plays its stored sequence whatever happens around it, and even a recording stored as changes from frame to frame applies stored differences instead of responding to anything. A program saved on a disk and the same program running differ in the same way. A dead brain keeps much of its anatomy and has lost the physiological activity that

anatomy used to support. Matching anatomy is therefore an incomplete comparison: the match is not exact, and a brain that matched a living one exactly, in its activity as well as its anatomy, would be running and alive. The account here is about that activity, including the interactions it supports.

A stored program lists operations. A running implementation holds the physical relations that carry those operations out. In a block description, running means that successive states have the causal organization of the process, including the relevant responses to possible inputs. Responses held in reserve are not enough. Tim Maudlin (1989) builds a machine whose responses to other inputs sit in idle parts while its actual run follows a fixed track; on the transitions view that run is a replay, since the track sets each next state and none of its states produces the next. No further computer or clock sits outside the block making it run. A recording is another structure within its own history, and resembling a process does not give the recording all of that process's causal relations. Both sides of this distinction are structures. Neither side is mathematics plus something that switches mathematics on.

4.1 Why a mind cannot run backward

A brain's thinking merges. Adding 1 and 1 gives 2, and so does adding 0 and 2. A brain's step takes its present state and its input to one next state, and many different earlier states lead to the same later one, with the difference carried off into the surroundings as heat (Landauer, 1961). A brain runs on about 20 watts, and all of it ends as heat. A computer built to keep every bit could retrace its steps, but a brain keeps no such record. Forward, a brain's next state is fixed by its present state and its input. Backward, they leave it open: $2 = x + y$ has more than one solution, and nothing in the brain's present state picks one. In a fully reversed history the choice is made by heat converging from the surroundings in

perfect coordination. A reversed brain would need its surroundings to supply, at each step, exactly the bits its forward steps discarded, in perfect coordination with its past. Forward inputs need no such tuning, so only the forward steps are produced by the brain.

Feeling therefore has a direction. Each felt transition is a step in which the mind produces its next state from its last, and that happens only forward. Memories are built by steps already taken, so a mind remembers the past and anticipates the future. Leonard Mlodinow and Todd Brun (2014) show that any memory that does not have to be fine-tuned to what it records points in the direction in which entropy, roughly a measure of disorder, rises. The arrow of time is not something a mind observes. It is what a mind is: a chain of states, each produced by the last.

5. Against premature closure

The transitions thesis sets a modeling requirement for sentience and a further reflective requirement for self-awareness. Rival theories draw the line elsewhere, and each comparison below shows what decides between them.

5.1 The hard problem of consciousness

David Chalmers (1995, 1996) asks why physical processes are accompanied by experience. The answer defended here is identity: the felt character is the organized transition itself. No separate process then has to attach feeling to an otherwise complete mind.

The hard problem of consciousness assumes its own conclusion: it takes feeling to be something other than structure, then finds no structure that explains it. The circle shows in the argument's form alone, so exposing it is an a priori proof, with no data needed. The assumption enters where the zombie argument moves from conceiving a structural copy without feeling to its possibility; conceiving water

without H2O does not make that possible either. Mind and body act on each other, so they share one structure, and any account of what feeling adds would itself be structure. Once the identity is in place, no bridge is owed: water needs no further law to make it H2O (Papineau, 2002).

Qualia theories (accounts of the felt qualities of experience), panpsychism (the view that all matter has some degree of mind), property dualism (the view that mental properties cannot be reduced to physical ones) and idealism (the view that reality is fundamentally mental) each face their own version of the same problem. Each must say what its extra ingredient is, how it combines into a subject and how it acts on the structure it is supposed to supplement, and each answer is more structure.

The identity proposal redirects investigation toward which structures constitute which experiences, and the transitions account gives its answer: running structures that model their world, producing each state from the last.

Evidence that a state is available for use is not evidence that it is felt. Ned Block (1995, 2002) separates phenomenal consciousness, the feeling itself, from access consciousness, a state's availability for report and reasoning. That split challenges this proposal directly. Availability is observable evidence; calling it the felt experience is a further claim. The proposal shares with Daniel Dennett (1991, 2005) and Keith Frankish (2016) an interest in explaining introspective judgments through mechanisms. It differs on one point: the felt character is a real part of the relevant organization. The identity argument of Section 1 supplies the first half of that position, and the modeling criterion of Section 2 the second.

6. AI consciousness as architectural question

Consciousness is felt transitions in self-modeling structure, so AI consciousness is an engineering question. Ask whether a system really models itself: does its

world-model include itself as an element, working inside its processing rather than printed as surface output?

A tool for reading a network's internals is not a test for feeling. A sparse autoencoder rewrites a network's internal activity as a set of learned features, one per recurring pattern. Researchers read those features and can test what each one does by switching it on or off. A feature that fires on descriptions of anxiety is not thereby an anxious experience. Adly Templeton and colleagues (2024) supply a method for investigating representations, not a validated test of sentience.

A useful test must discriminate between rival mechanisms. The proposed experiment compares two systems that score the same on perceptual task accuracy and differ in one independently specified mechanism: the one that tracks their own errors and uncertainty. Selectively perturb that mechanism and compare effects with perturbations matched for general performance loss. Measure confidence calibration (how well a system's confidence matches its actual accuracy), error detection and adaptation to unfamiliar failures. A selective loss of those capacities would support a functional role for self-modeling; it would not alone establish felt experience.

Mathematical monism does not prohibit experience in artificial systems. What a system is built from decides which causal relations it has. Evidence about architecture can inform welfare research without being advertised as proof that a deployed system feels.

6.1 Activity that keeps running

A brain's activity keeps running while it works. A pattern that carries what the mind is doing keeps firing and feeds back on itself through loops at many scales, updating in place as the work goes on, and what a brain is aware of at a moment goes with activity that is sustained and shared widely across the cortex, the

brain's outer layer (Dehaene et al., 2011). Anesthesia can leave local circuits working while it disrupts that integration, and experience goes with the integration (Alkire, Hudetz and Tononi, 2008).

A large language model, as built today, works differently. For each token, a word or piece of a word, its activity sweeps once through its layers in one direction and stops, and nothing inside a sweep loops back (Vaswani et al., 2017). What carries over to the next token is the token itself and a cache of the activity that earlier sweeps stored, which the next sweep reads. While the model searches or reasons, the continuity lies in the text it writes and in that cache. It consults a record of its earlier states and does not keep them running.

On the transitions view, each sweep is a run, a short chain of states producing states, dozens of layers deep, and what links the sweeps is a stored record that each new sweep reads and builds on. A self-model that forms inside a sweep lasts that sweep, and the next sweep meets it as stored numbers. Brains also store memories in their connections and read them back. The difference lies in the working model of the moment, which in a brain keeps running and in a current language model is rebuilt from the record at every token.

Borjan Milinkovic and Jaan Aru (2025) describe three features of biological computation that current artificial systems lack: it is hybrid, mixing discrete events with continuous dynamics; it is scale-inseparable, with no clean line between the algorithm and the tissue that runs it; and it is metabolically grounded, shaped by the brain's energy limits. The transitions view adds a fourth, more basic for feeling: a brain's model of what it is doing keeps running while it works. Energy limits are a difference of degree, since a data center also runs within a power budget, though a far looser one than the brain's twenty watts.

How an artificial system could feel, on this view, is a question of architecture. A system whose model of itself keeps running while it works, updating in place and

feeding back into its own processing, would meet the condition, whatever it is made of. Digital steps are enough, since on the view taken here every process runs in steps: a recurrent loop simulated step by step on today's processors would run in the sense used here, its updates playing the part that sustained electrical activity plays in a brain. Spiking (signaling in brief pulses, as neurons do), asynchronous (running without a shared clock) or analog hardware would mainly change the energy and speed of such a loop. Integrated information theory disagrees, since it scores the physical hardware and not the simulated loop (Tononi et al., 2016). The test above applies to such a system without change.

7. Objections and evidence from comparative cognition

First, self-modeling may occur without experience. Another structural theory could identify experience with a different organization. Answering that objection takes comparative argument and evidence; restating the identity under test will not do it.

Second, feedback makes the proposal trivial. Feedback, recursion (a procedure defined in terms of itself) and representation of one's own representational activity are distinct. A candidate self-model has to be shown tracking the system's own processing, and it has to support interventions that separate its contribution from ordinary task computation.

Third, many animals show little evidence of reflective self-awareness. The sentience proposal must be assessed separately from the reflective proposal. Evidence about pain, learning and flexible behavior in fish, octopuses or insects cannot be redescribed as evidence of deep self-modeling without additional argument.

Depth of modeling may not be the only axis along which animal minds differ. Jonathan Birch, Alexandra Schnell and Nicola Clayton (2020) propose five dimensions of animal consciousness. Their multidimensional approach offers a

useful alternative to a single ladder of modeling depth. The present proposal needs to explain its relationship to those dimensions.

Map the internal features that activate on the system's own representational activity. Characterize how activity changes in time around outputs the system identifies as state-relevant. Design intervention experiments that ablate (switch off) self-referential computation. Differences between located sensation, emotion and reflection are phenomena for the proposal to explain. They should not be assumed to form a single ordering by modeling depth.

8. Conclusion

Feeling lives in the transitions of a modeling mind, because a mind is its complete structure and experience is that structure at work. Simple laws grow minds in causal steps, each state produced from the last, and the steps are the feeling. A photograph, a description and a recording produce nothing and feel nothing. Feeling runs only forward, because a thought merges many states into one and only the forward step is produced by the mind itself. That direction is the felt arrow of time.

Defining experience as non-structural settles the question before it is asked, since no added structural detail can close a gap built in by definition. Mathematical monism rejects that definition. What remains is sorting systems by their organization: which ones model their world, which ones also model themselves, and which only store or replay states. The sorting applies to brains and artificial systems alike. A quantum mechanism of consciousness, if established, would also be a structural account; the central argument does not depend on it.

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